

John Thickstun

+1 (315) 601-5709

jthickstun@cornell.edu

<https://johnthickstun.com/>

Current Position

- **Cornell University** Ithaca, NY
Assistant Professor - Department of Computer Science July 2024 - Present

Education

- **Stanford University** Palo Alto, CA
Postdoctoral Scholar - Stanford Artificial Intelligence Laboratory September 2021 - June 2024
– Advised by Percy Liang (joint Computer Science & Statistics).
- **University of Washington** Seattle, WA
Ph.D. in Computer Science and Engineering August 2021
– Advised by Sham M. Kakade (joint Computer Science & Statistics) and Zaid Harchaoui (Statistics).
– Doctoral Committee: Sham M. Kakade, Zaid Harchaoui, Noah A. Smith, Sewoong Oh, Lalit Jain.
– Dissertation: Leveraging Generative Models for Music and Signal Processing.
- **University of Washington** Seattle, WA
M.Sc. in Computer Science and Engineering December 2017
– Coursework in Optimization, Algorithms, Learning Theory, Information Theory, Reinforcement Learning.
- **Brown University** Providence, RI
Sc.B. Magna cum Laude with Honors in Applied Mathematics May 2013
– Advised by Eugene Charniak (Computer Science) and Björn Sandstede (Applied Mathematics).
– Coursework in Machine Learning, Computer Vision, Natural Language Processing, Probability Theory, Stochastic Processes, Real and Complex Analysis, Operator Theory.

Major Grants and Awards

- Anonymous Donor Gift for “AI and Music”: \$100,000 (2025).
- Sound Patrol Research Gift: \$250,000 (2025).
- ACL Outstanding Paper Award: 39 / 3872 paper submissions (2023).
- Stanford HAI Google Cloud Credits Grant: \$15,000 (2022).
- Neurips Outstanding Paper Award: 6 / 9122 paper submissions (2021).
- Qualcomm Innovation Fellowship: \$100,000 (2020).
- NSF Graduate Research Fellowship: \$138,000 (2017-2019).
- Brown University Distinguished Senior Thesis Prize (university-wide award, 2013).
- Sigma Xi Scientific Honor Society (2013).

Professional Experience

- **Sound Patrol** - Generative Music Technology Los Angeles, CA
Head of Research December 2023 - Present
– Technical leadership on training foundation models for music information retrieval and copyright protection.
– Hiring and mentorship of research engineers.
- **Clear Ventures** - Technology Venture Capital Palo Alto, CA
DeepTech Fellow September 2022 - May 2023
- **Panjandrum.ai** - Virtual Music Avatars Seattle, WA
Advisor October 2019 - May 2021

- Advice and technical strategy for applications of machine learning to music and visual production.

- **University of Washington** Seattle, WA
Graduate Researcher - Computer Science and Engineering *September 2015 - August 2021*
 - Built a track-record of machine learning research with publications at ICML, Neurips, ICLR, ICASSP, ISMIR.
 - Created the current best music transcription model in the MIREX Multi-F0 Challenge.
 - Designed, built, and administered a GPU computing cluster to support two research groups.
- **Amazon** Seattle, WA
Applied Science Intern - Amazon Music Machine Learning *June 2019 – Aug. 2019*
 - Hosts: Ted Sandler, Ben London.
 - Built a recommendation model using deep contextual bandits to sequence tracks on Amazon Music stations.
 - Used counterfactual risk minimization to train models off-policy from logged user interaction data.
- **Bracebridge Capital - Fixed Income Arbitrage Hedge Fund** Boston, MA
Quantitative Developer (Lead Developer) - Quantitative Research *July. 2013 – May 2015*
 - Led and mentored a team of three software engineers developing C++ software infrastructure.
 - Maintained models governing a billion dollar asset backed structured product portfolio.
 - Organized front-office data acquisition, coordinating between research, vendors, the trading floor, and IT.
 - Built in-house models of implied volatilities and sensitivities, with applications to rates products.
- **Bracebridge Capital** Boston, MA
Summer Analyst - Quantitative Research *June 2012 – Aug. 2012*
 - Rebuilt a legacy Excel product model using modern technologies: C#, Postgres, and JavaScript.
 - Completed a quant training course on valuation and risk models, with a focus on fixed income products.
- **Sirius Software - Database Vendor** Cambridge, MA
Software Engineer - Systems Software Engineering *Jan. 2008 – May 2012*
 - Wrote compiler extensions for a database bytecode query language used by 1000+ developers worldwide.
 - Developed core system libraries for reporting, parsing, and web, supporting applications with millions of users.
 - Delivered international on-site programmer training and product demonstrations.
 - Linux systems administration: DNS, backups, software updates, security, documentation wiki.

Research Mentoring

PhD Students

- Michael Freeman, Cornell University.
 - Michael and I have been working together since Fall 2025.
- Saebyeol Shin, Cornell University.
 - Saebyeol and I have been working together since Summer 2025.
- Kimia Kazemian, Cornell University.
 - Kimia and I have been working together since Spring 2025.
- Subham Sahoo, Cornell University.
 - Subham and I have been working together since Spring 2025.
- Zhihan Yang, Cornell University.
 - Zhihan and I have been working together since Fall 2024.
- Rohith Kuditipudi, Stanford University.
 - Rohith and I co-authored a paper appearing in TMLR 2024, and a post on the Stanford CRFM blog.
- Megha Srivastava, Stanford University.
 - Megha and I co-authored a paper appearing in TMLR 2023, and a post on the Stanford CRFM blog.
- Xiang Lisa Li, Stanford University.

- Lisa and I co-authored a paper appearing at Neurips 2022.
- Vivek Jayaram, University of Washington.
 - Vivek and I co-authored papers appearing at ICML 2020, ICML 2021, and Neurips 2025.

Masters Students

- Nic Becker, Stanford University, Class of 2025.
 - Nic and I co-authored a workshop paper appearing in GenAICHI 2024.

Undergraduate Students

- Daniel Lin, Cornell University, Class of 2027.
 - Daniel and I co-authored a workshop paper together.
- Katherine Li, Stanford University, Class of 2025.
 - [CURIS](#) summer research student, 2023.
- Jasmine Gonzales, Stanford University, Class of 2025.
 - [CURIS](#) summer research student, 2023.
- Harsh Verma, University of Washington, Class of 2019.
 - Harsh and I co-authored a paper appearing at ISMIR 2019.

Teaching

- Instructor. Cornell CS 3780/5780: [Introduction to Machine Learning](#) (Fall, 2025).
 - Taught a class of 464 students.
 - Managed a team of 40 teaching assistants to create homeworks, hold office hours, grade assignments.
- Instructor. Cornell CS 6785: Deep Probabilistic and Generative Models (Spring, 2025).
 - Taught 40 students at the Cornell and Cornell Tech campuses.
- Instructor. Cornell CS 3780/5780: [Introduction to Machine Learning](#) (Fall, 2024).
 - Taught a class of 381 students (with Sarah Dean and Thorsten Joachims).
 - Developed new lectures and homework on modern neural networks: Transformers, Large Language Models.
 - Managed a team of 35 teaching assistants to create homeworks, hold office hours, grade assignments.
- Predoctoral Instructor (Instructor of Record). UW CSE 599i: [Generative Models](#) (Fall, 2020).
 - Taught a class of 30 students.
 - Created a new course offering covering advances in generative modeling from 2010-2020.
 - Developed course materials from scratch including lecture notes, slides, and homework.
 - Top-decile teaching reviews: 4.9/5.0 overall course quality with a 53% response rate (16/30 students).
- Teaching Assistant (University of Washington)
 - CSE547: Machine Learning for Big Data (Spring, 2016).
 - CSE546: Machine Learning (Autumn, 2015).
- Guest Lecturer (University of Washington)
 - STAT558: Statistical Machine Learning for Data Scientists (Spring 2020).
 - CSE490: An Introduction to Deep Learning (Autumn, 2018).
 - STAT558: Statistical Machine Learning for Data Scientists (Spring 2018).
 - CSE546: Machine Learning (Spring 2018).
 - CSE547: Machine Learning for Big Data (Spring, 2016).

Professional Service

- Program Committees:
 - Area Chair, Conference on Language Modeling, 2025, 2026.
 - Area Chair, International Conference on Learning Representations, 2025, 2026.
 - Meta Reviewer, International Society for Music Information Retrieval, 2024.

- Grant Reviewer:
 - Schmidt Sciences: AI Safety, 2025.
- Journal Reviewer:
 - Computational Linguistics, 2025.
 - IEEE Transactions on Pattern Analysis and Machine Intelligence 2024.
 - ACM Computing Surveys 2023.
 - Journal of Creative Music Systems 2022, 2023.
 - IEEE Signal Processing Letters 2022.
 - Transactions of the International Symposium on Music Information Retrieval 2021, 2022.
- Conference Reviewer:
 - Conference on Language Modeling 2024.
 - International Conference on Learning Representations 2022, 2023, 2024.
 - International Conference on Machine Learning 2018, 2021, 2022.
 - Advances in Neural Information Processing Systems 2016, 2020, 2021, 2022, 2023, 2024.
- Workshop Reviewer:
 - NeurIPS Workshop on AI for Music, 2025.
 - ICLR Workshop on Mathematical and Empirical Understanding of Foundation Models, 2024.
 - NeurIPS Workshop on Machine Learning Safety, 2022.
- Stanford CS [Undergraduate Mentoring](#), 2021-2022.
 - This program provides early research mentoring to undergraduate students from underrepresented groups.
 - Met regularly with my mentee during the 2021-2022 academic year.
- UW CSE Application Reader, PhD Admissions: 2018, 2019, 2020.
 - Screened and reviewed 30-50 PhD applications annually for the UW CSE Machine Learning group.
- UW CSE Machine Learning Graduate Student Recruiting Activities Coordinator: 2018, 2019.
 - Planned and organized on-campus recruiting events and activities for 100+ current and prospective students.
- UW CSE Graduate Social Co-Chair, 2017.
 - Organized weekly student social events for the Allen School graduate student community.
 - Worked to create inclusive activities that are accessible and appealing to our diverse community of students.
- Co-founder and organizer of the UW Machine Learning and Optimization Reading Group.
 - Organized and scheduled speakers for a weekly seminar for 5 years (2015-2020).
 - In 2020, this seminar grew into the regular meeting of [ADSI](#)/ [IFDS](#), funded by an NSF Tripods grant.

Outreach

- [Frontier AI and preparing the Next Leap](#)
 - October 22, 2025: panelist at TEDAI, alongside Rafael Rafailov and Llion Jones (moderated by Jørn Lyseggen).
 - Topic: a deep dive into post-transformer architectures, multi-modal foundation models, and AI agents. What's coming after today's dominant systems, and how it will reshape industries.
- Priorities for Federal Funding in AI
 - May 29, 2025: Authored a response to a federal request for information regarding development of the 2025 National AI R&D Strategic Plan.
- [Exploring the Ethics of AI through Narrative](#)
 - April 03, 2025: A retrospective on Stories for the Future: a workshop convened between filmmakers and AI researchers at Stanford University.
- [Securing America's AI Future](#)
 - March 14, 2025: Co-authored a response to an information request for the White House's AI Action Plan.
- [The IFML Podcast](#): Meet the research team behind MAUVE
 - May 22, 2024: The inaugural episode of the [IFML](#) podcast (Institute for Foundations of Machine Learning).

- California Assembly Briefing: What's Real in the Technology Combating Deepfakes?
 - May 20, 2024: An information-gathering panel discussion for staffers to the California State Assembly.
 - I was invited as a panelist to share my expertise on the topic of text watermarking.
- [Stories for the Future Workshop](#)
 - May 17, 2024: A gathering of 11 sci-fi filmmakers and AI researchers for a day-and-a-half experiment in fostering new narratives about AI.
- [Press Play](#): Carol Reilly and the Robots, April 6 & 7, 2024
 - I premiered a musical composition co-composed with the Anticipatory Music Transformer.
 - In collaboration with Carol Reilly and performers from the San Francisco Symphony Orchestra.
- [Anticipatory Music Transformer: A Composer's Helper](#)
 - December 4, 2023: a brief discussion of the Anticipatory Music Transformer with [Stanford HAI](#) on YouTube.
- Stanford Student-Applicant Support Program ([SASP](#)) 2022.
 - Provided feedback on PhD applications to prospective Stanford students from under-represented groups.
- UW CSE Pre-Application Mentorship Service ([PAMS](#)) 2021.
 - Mentored prospective applicants to the UW PhD program from historically marginalized groups.
- Academic Research Panels:
 - June 2021: Howard University Karsh STEM Scholars Research Panel for Incoming First-Year Students
 - March 2021: UW CSE PhD Student Experience Panel for Admitted Graduate Students
 - February 2020: CSE 142 Careers in Research Panel for First-Year Computer Science Students
 - May 2018: ACM Research Night for UW Undergraduate Students

Publications and Preprints

Theses

1. Leveraging Generative Models for Music and Signal Processing.
University of Washington Dissertation.
Paul G. Allen School of Computer Science & Engineering, 2021.
[John Thickstun](#).
2. Statistical Inference on Music with Applications to the Transcription Problem.
Brown University Senior Thesis.
Department of Applied Mathematics, 2013.
Brown University **Distinguished Senior Thesis Award** (university-wide award).
[John Thickstun](#).

Journal Publications

1. Robust Distortion-free Watermarks for Language Models.
Transactions on Machine Learning Research (**TMLR**) 2024.
Rohith Kudithipudi, [John Thickstun](#), Tatsunori Hashimoto, Percy Liang.
2. Anticipatory Music Transformer.
Transactions on Machine Learning Research (**TMLR**) 2024.
[John Thickstun](#), David Hall, Chris Donahue, Percy Liang.
3. MAUVE Scores for Generative Models: Theory and Practice.
Journal of Machine Learning Research (**JMLR**) 2023.
Krishna Pillutla, Lang Liu, [John Thickstun](#), Sean Welleck, Swabha Swayamdipta, Rowan Zellers, Sewoong Oh, Yejin Choi, Zaid Harchaoui.
4. Evaluating Human-Language Model Interaction.
Transactions on Machine Learning Research (**TMLR**) 2023.
Mina Lee, Megha Srivastava, Amelia Hardy, [John Thickstun](#), Esin Durmus, Ashwin Paranjape, Ines Gerard-Ursin, Xiang Lisa Li, Faisal Ladhak, Frieda Rong, Rose E. Wang, Minae Kwon, Joon Sung Park, Hancheng Cao, Tony Lee, Rishi Bommasani, Michael Bernstein, Percy Liang.

Conference Publications

1. Linearly Constrained Diffusion Implicit Models.
In Advances in Neural Information Processing Systems (**Neurips**) 2025.
Vivek Jayaram, Ira Kemelmacher-Shlizerman, Steve Seitz, [John Thickstun](#).
2. Aligning Text-to-Music Evaluation with Human Preferences.
In International Symposium on Music Information Retrieval (**ISMIR**) 2025.
Yichen Huang, Zachary Novack, Koichi Saito, Jiatong Shi, Shinji Watanabe, Yuki Mitsufuji, [John Thickstun](#), Chris Donahue.
3. Backpack Language Models.
In Annual Meeting of the Association for Computational Linguistics (**ACL**) 2023.
Outstanding Paper Award: 39 / 3872 paper submissions.
John Hewitt, [John Thickstun](#), Christopher D. Manning, Percy Liang.
4. Melody Transcription via Generative Pre-Training.
In International Symposium on Music Information Retrieval (**ISMIR**) 2022.
Chris Donahue, [John Thickstun](#), Percy Liang.
5. Diffusion-LM Improves Controllable Text Generation.
In Advances in Neural Information Processing Systems (**Neurips**) 2022.
Selected for **Oral Presentation**.
Xiang Lisa Li, [John Thickstun](#), Ishaan Gulrajani, Percy Liang, Tatsunori B. Hashimoto.
6. MAUVE: Measuring the Gap Between Neural Text and Human Text using Divergence Frontiers.
In Advances in Neural Information Processing Systems (**Neurips**) 2021.
Outstanding Paper Award: 6 / 9122 paper submissions.
Krishna Pillutla, Swabha Swayamdipta, Rowan Zellers, [John Thickstun](#), Sean Welleck, Yejin Choi, Zaid Harchaoui.
7. Parallel and Flexible Sampling from Autoregressive Models via Langevin Dynamics.
In International Conference on Machine Learning (**ICML**) 2021.
Vivek Jayaram*, [John Thickstun](#)* (*equal contribution).
8. Faster Policy Learning with Continuous-Time Gradients.
In Learning for Dynamics & Control (**L4DC**) 2021.
Samuel Ainsworth, Kendall Lowrey, [John Thickstun](#), Zaid Harchaoui, Siddhartha Srinivasa.
9. An Information Bottleneck Approach for Controlling Conciseness in Rationale Extraction.
In Empirical Methods in Natural Language Processing (**EMNLP**) 2020.
Bhargavi Paranjape, Mandar Joshi, [John Thickstun](#), Hannaneh Hajishirzi, Luke Zettlemoyer.
10. Source Separation with Deep Generative Priors.
In International Conference on Machine Learning (**ICML**) 2020.
Vivek Jayaram*, [John Thickstun](#)* (*equal contribution).
11. Convolutional Composer Classification.
In International Symposium on Music Information Retrieval (**ISMIR**) 2019.
Harsh Verma, [John Thickstun](#).
12. Coupled Recurrent Models for Polyphonic Music Composition.
In International Symposium on Music Information Retrieval (**ISMIR**) 2019.
[John Thickstun](#), Zaid Harchaoui, Dean P. Foster, Sham M. Kakade.
13. Invariances and Data Augmentation for Supervised Music Transcription.
In International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**) 2018.
Selected for **Oral Presentation**.
[John Thickstun](#), Zaid Harchaoui, Dean P. Foster, Sham M. Kakade.
14. [MusicNet](#): Learning Features of Music from Scratch.
In International Conference on Learning Representations (**ICLR**) 2017.
[John Thickstun](#), Zaid Harchaoui, Sham M. Kakade.

Workshop Papers

1. The Reliability Gap in Agentic Evidence Verification for Materials Science.
In ICLR Workshops on Agents in the Wild: Safety, Security (AIWILD) and Beyond and Foundation Models for Science (FM4Science), 2026. Albert Gong, James J. Kim, Anmol Kabra, Aaditya Panigrahi, Jiashuo Wang, Arjun B. Mulchandani, Michael Freeman, Fatmagul Katmer, Joshua Peters Wakefield, Linxi Zhao, Chao Wan, Akanksha Sarkar, Yoav Artzi, Leslie M. Schoop, [John Thickstun](#), Kilian Q. Weinberger, Eun-Ah Kim, Peter I. Frazier, Jennifer J. Sun.
2. Esoteric Language Models: Bridging Autoregressive and Masked Diffusion Paradigms.
In ICLR Workshop on Multimodal Intelligence: Next Token Prediction & Beyond, 2026.
Subham Sekhar Sahoo, Zhihan Yang, Yash Akhauri, Johnna Liu, Deepansha Singh, Zhoujun Cheng, Zhengzhong Liu, Eric P. Xing, [John Thickstun](#), Arash Vahdat.
3. Robust Neural Audio Fingerprinting using Music Foundation Models.
In NeurIPS Workshop on AI for Music, 2025.
Shubhr Singh, Kiran Bhat, Benjamin Resnick, Xavier Riley, [John Thickstun](#), Walter De Brouwer.
4. Hookpad Aria: A Copilot for Songwriters.
In International Symposium on Music Information Retrieval: Late-Breaking Demos (**ISMIR LBD**), 2024.
Chris Donahue, Shih-Lun Wu, Yewon Kim, Dave Carlton, Ryan Miyakawa, [John Thickstun](#).
5. Designing Live Human-AI Collaboration for Musical Improvisation.
In Generative AI and HCI Workshop at CHI (**GenAICHI**), 2024.
Nic Becker, Ryan Louie, [John Thickstun](#), Percy Liang.
6. Frequency Domain Convolutions for Multiple F0 Estimation.
MIREX Abstract (Technical Report) 2017.
[John Thickstun](#), Zaid Harchaoui, Dean P. Foster, Sham M. Kakade.

Pre-print Reports

1. Scaling Beyond Masked Diffusion Language Models.
Under Review, 2026. ArXiv Preprint 2602.15014
Subham Sekhar Sahoo, Jean-Marie Lemerrier, Zhihan Yang, Justin Deschenaux, Jingyu Liu, [John Thickstun](#), Ante Jukic.
2. Factual and Musical Evaluation Metrics for Music Language Models.
Under Review, 2025. ArXiv Preprint 2511.05550
Daniel Chenyu Lin, Michael Freeman, [John Thickstun](#).
3. Benchmark Datasets for Lead-Lag Forecasting on Social Platforms.
Under Review, 2025. ArXiv Preprint 2511.03877
Kimia Kazemian, Zhenzhen Liu, Yangfanyu Yang, Katie Z. Luo, Shuhan Gu, Moyun Du, Jack Jansons, Kilian Q. Weinberger, [John Thickstun](#), Yian Yin, Sarah Dean.
4. Esoteric Language Models.
Under Review, 2025. ArXiv Preprint 2506.01928
Subham Sekhar Sahoo, Zhihan Yang, Yash Akhauri, Johnna Liu, Deepansha Singh, Zhoujun Cheng, Zhengzhong Liu, Eric Xing, [John Thickstun](#), Arash Vahdat.
5. From Forensics to Ecosystems: Rethinking Watermarks for Generative AI Oversight.
Under Review, 2025.
Daniel Susser [John Thickstun](#), Gili Vidan.

Invited Talks

- [Controlling Artificial Intelligence](#)
- TED AI - San Francisco, CA - 10/21/2025

- Robust Distortion-free Watermarks for Language Models
 - UCLA - Los Angeles, CA - 2/18/2026
 - Cornell AI Seminar - Ithaca, NY - 9/6/2024
 - Google Algorithms Seminar - New York, NY - 7/2/2024
 - Google Differential Privacy Seminar - Mountain View, CA - 7/12/2023
- Mechanisms and Metaphors for Interactions with AI
 - Stories for the Future Workshop - Stanford University - 5/17/2024
- Control Mechanisms for Generative Models
 - Cornell - Ithaca, NY - 4/9/2024
 - Cornell Tech - New York City, NY - 4/8/2024
 - Boston University - Boston, MA - 3/25/2024
 - UC San Diego - La Jolla, CA - 2/28/2024
- Anticipation and the Anticipatory Music Transformer
 - Grinnell College - Grinnell, IA - 05/02/2024
 - Spotify - New York, NY - 04/24/2024
 - University of Wisconsin - Madison, WI - 11/10/2023
 - MIT - Cambridge, MA - 10/30/2023
 - Harvard - Cambridge, MA - 10/27/2023
 - UC San Diego - La Jolla, CA - 8/23/2023
- Advancing the Capabilities of Generative AI: Two Vignettes
 - Radix Trading - Chicago, IL - 6/8/2023
- Generative Models for Controllable Content Creation
 - CLEAR Ventures - Palo Alto, CA - 10/24/2022
 - Meta - New York, NY - 10/18/2022
- Audio Source Separation with Deep Generative Priors
 - Mila - Quebec, Canada - 8/19/2022
- Classifier-Guided Controllable Text Generation with Diffusion-LM
 - AI2 - Seattle, WA - 8/3/2022
- Generative Modeling of Classical Western Music
 - SAIL - Stanford University - 12/4/2020
- Source Separation with Deep Generative Priors
 - ICML - Vienna, Austria - 7/14/2020
- Robust Generative Modeling in Diverse Modalities
 - CSE Colloquium - UW Seattle - 10/31/2019
- Automatic Music Transcription
 - CS Department - Brown University - 5/1/2013
- Introducing the Janus XmlParser
 - Sirius User Group - St. Louis, MO - 5/2/2010
- Tokenization and Collection Objects
 - Centrelink - Canberra, Australia - 3/23/2010

Media Coverage

Generative AI

- Engadget - [X's 'open source' algorithm isn't a win for transparency, researchers say.](#)
- Yahoo Finance - [Silicon Valley is going all in on 'superintelligent' AI, and there's plenty of hype.](#)
- Associated Press - [OpenAI launches GPT-5, a potential barometer for whether AI hype is justified.](#)
- CNET - [Gen AI's Accuracy Problems Aren't Going Away Anytime Soon, Researchers Say.](#)

The Anticipatory Music Transformer

- San Francisco Chronicle - [Beethoven meets AI as music enters a new technological moment.](#)
- The Baltimore Sun - [BSO partners with tech firm to perform music composed by artificial intelligence .](#)
- The AudioCipher Blog - [Hooktheory Aria: The New AI MIDI Generation Tool in HookPad.](#)

Watermarking Language Models

- Axios - [Google DeepMind open sources its AI text watermarking tool.](#)
- Brookings - [Detecting AI fingerprints: A guide to watermarking and beyond.](#)
- Wall Street Journal - [There's a Tool to Catch Students Cheating With ChatGPT. OpenAI Hasn't Released It.](#)

MusicNet

- TechCrunch - [MusicNet aims to give machine learning algorithms a taste for Beethoven.](#)
- The Times of London - [Bach to the future: computer will finish composer's work.](#)
- A Tempo with Rachel Katz (WWFM Radio) - [Computers and music.](#)
- CNET - [Bach to the future: AI, meet classical music.](#)
- CIFAR - [Learning algorithms find a new music teacher.](#)
- RouteNote - [How do you advance machine learning? Teach them Beethoven and Bach.](#)
- Technical.ly - [AI can't replace the violinist's hands, but it can generate musical compositions for the BSO.](#)
- UW Today - [What makes Bach sound like Bach? New dataset teaches algorithms classical music.](#)

Open Source Contributions

- Stanford CRFM Mistral
 - <https://github.com/stanford-crfm/mistral>
 - A framework for replicable training of GPT-2 scale Transformer models.
 - I contributed abstractions for training models over non-text datasets including, for example, music.
- HuggingFace Datasets
 - <https://github.com/huggingface/datasets/>
 - A Python library of natural language processing datasets and utilities.
 - I contributed model evaluation metrics that implement the MAUVE algorithm for measuring the quality of machine-generated text.
- FIFE Engine
 - <https://github.com/fifengine/fifengine>
 - A multi-platform isometric game engine written in C++ with Python bindings for scripting.
 - I was an early core contributor from 2007-2008 with 241 commits touching all aspects of the engine.

Blog Posts

- [Bowers Faculty Outline Priorities for Federal Funding in AI](#) - 2025/06/02
 - [Cornell CS](#) - Newsfeed for Cornell Computer Science
- [Co-Composition with an Anticipatory Music Transformer](#) - 2024/08/06
 - [The CRFM Blog](#) - Stanford Center for Research on Foundation Models
- [A Closer Look at Human-LM Interactions in Information-Seeking Contexts](#) - 2023/10/11
 - [The CRFM Blog](#) - Stanford Center for Research on Foundation Models
- [Robust Distortion-free Watermarks for Language Models](#) - 2023/07/30
 - [The CRFM Blog](#) - Stanford Center for Research on Foundation Models
- [Anticipatory Music Transformer: A Controllable Infilling Model for Music](#) - 2023/06/16
 - [The CRFM Blog](#) - Stanford Center for Research on Foundation Models
- [Acoustic Models for Music Transcription](#) - 2018/03/15
 - [ADSI](#) - Algorithmic Foundations of Data Science Institute